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MZUMBE UNIVERSITY
COMPUTING SCIENCE STUDIES

CSS 213: SYSTEM ANALYSIS & DESIGN
BSc. ITS-II, ICTB-II, ICTM-II & MICT-EDU-II
TEST 1: SEM_1 2020/2021

Time: 1 HR

QUESTION ONE

1. Why a prototype is important in system analysis and design? **(2 Marks)**
2. What is the difference between system **VALIDATION** and system **VERIFICATION** **(2 Marks)**
3. Why should the system request be created by a business person as opposed to an IS professional? **(2Marks)**

- Better Idea of business needs and values that will come to the organization.

QUESTION TWO

1. Explain the use of a flowchart and Algorithm in system analysis & design **(2 Marks)**
2. Draw a flowchart to determine the largest of three numbers **(4 Marks)**
3. Algorithm to find the greatest of three numbers using an additional variable MAX **(4 Marks)**

Prepared by Dr. S.B Kisanjara

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MZUMBE UNIVERSITY
(CHUO KIKUU MZUMBE)

Faculty of Science and Technology

End of Semester I Examinations: March, 2020

Course Code:	CSS 213
Course Title:	System Analysis and Design
Programme:	ITS II, ICTM II, ICTB II and MICT-EDU-II
Date:	10-03-2020
Time:	1300 - 1600
Venue:	NAH

INSTRUCTIONS

1. There are **FIVE (5)** questions in this paper
2. Attempt **Question One(1)** and **Any Three (3)** Questions
3. Each Questions Carries **20%** except Question 1 which carries **40%**

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QUESTION ONE (40 MARKS)

Student Resume Management System (SRMS)

Pretend that the career services office at your university wants to develop a system that collects student resumes and makes them available to students and recruiters over the Web. Students should be able to input their resume information into a standard resume template. The information then is presented in a resume format, and it also is placed in a database that can be queried through an online search form. You have been placed in charge of the project.

i. What are the six (6) general skills all project team members should have? **(6 Marks)**

ii. How long do you think it would take for you and three other students to complete the project?. Provide support for the plan schedule that you propose. **(6 Marks)**

iii. Create a work plan listing the tasks that will need to be completed to meet the project's objectives **(10 Marks)**

iv. Create a **Gantt chart** and a **PERT chart** of the proposed system **(18 Marks)**

QUESTION TWO (20 MARKS)

i. What are the FOUR key principles or values of Agile software development? **(4 Marks)**

ii. Compare and contrast phases, steps, techniques, and deliverables.. **(6 Marks)**

iii. During Structured Analysis, various tools and techniques are used for system development. List and describe in brief , five (5) techniques we have learned in class sessions. **(10 Marks)**

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QUESTION THREE

(20 MARKS)

i. Briefly describe what do **Verification** and **Validation** processes does in the practitioner of system **(6 Marks)**

ii. What is the difference between a methodology and a work plan? How are the two terms related? **(8 Marks)**

iii. Distinguish between logical process models and physical process models. **(6 Marks)**

QUESTION FOUR

(20 MARKS)

i. Write short notes on "Documentation control". In your explanation state what are its advantages and types **(6 Marks)**

ii. What are some problems associated with using packaged software? How can these problems be minimized? **(6 Marks)**

iii. Discuss how the analyst knows how to stop decomposing the process model into more and more levels of detail when designing a DFD's diagram of the system **(8 Marks)**

QUESTION FIVE

(20 MARKS)

i. What do CASE used for in SAD? **(2 Marks)**

ii. Suppose that a process on a DFD is numbered 4.3.2. What level diagram contains this process? What is this process's parent process? **(6 Marks)**

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iii. Briefly describe the differences that exist between different phases of system development life cycle.

(12 marks)

(12 Marks)

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Course Code:	CSS 213
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Date:	30/3/2021
Time:	1600 – 1900 HRS
Venue:	OC

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QUESTION ONE (25 MARKS)

a) What is the role of system analyst? **(1Marks)**

b) Why system analysis and design is important in software development ? **(6 Marks)**

c) With example explain where do system analysis and design project idea come from? **(6 Marks)**

d) System modeling has now come to mean representing a system using some kind of graphical notation, which is now almost always based on notations in the Unified Modeling Language (UML). Why system modeling is important to system analyst? **(12 Marks)**

QUESTION TWO (25 MARKS)

a) There are several types of UML diagrams. Mention and explain only five (5) UML diagrams you know? **(10 Marks)**

b) What is requirement determination? List three (3) activities involved **(6 Marks)**

c) Why it is important to carryout feasibility study in system analysis and design project? **(9 Marks)**

QUESTION THREE (25 MARKS)

a) What is the importance of use case diagram? **(6 Marks)**

b) Why it is important to develop a prototype in system design process before the final output/product? **(10 Marks)**

c) Testing is the process or activity that checks the functionality and correctness of software according to specified user requirements in order to improve the quality and reliability of system. It is an expensive, time consuming, and critical approach in system development which requires proper planning of overall testing process and set of rules. Briefly explain nine (9) rules for system guiding testing process? **(9 Marks)**

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QUESTION FOUR **(25 MARKS)**

a) Define the following terms (1) Algorithm (2) Pseudo code as applied in system analysis and design **(2 Marks)**

b) What is the main essence of performing Pseudo code? **(2 Marks)**

c) Write pseudo code that tells a user that "the number they entered is not a 5 or a 6"? **(10 Marks)**

d) Design an algorithm and the corresponding flowchart for "finding the sum of n numbers"? **(11 Marks)**

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SYSTEM ANALYSIS AND DESIGN CSS_213 TEST II
BSc. ITS-II, ICTB-II, ICTB-II & MICTEDU-II
SEMESTER I_2021/2022 ACADEMIC YEAR

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- Time: 40 Mins.
- Answer All Questions in this Paper briefly with at most six (6) lines per answer:

1. What is the Unified Modeling Language (UML)? How does it support the approach to systems Development? (5 Marks)
2. System modeling has now come to mean representing a system using some kind of graphical notation, which is now almost always based on notations in the Unified Modeling Language (UML). Why system modeling is important to system analyst? (5 Marks)
3. Why testing is very crucial part in development of new system? (5 Marks)

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CSS 224.

BSc. IIS II.

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Briefly answer the

1. What do you Understand the followings:- Inheritance: data encapsulation and polymorphism (3Marks)
2. What is the main foaus of OOD (3Marks
3. Why generalization relationship is important in Class diagram? (3 Marks)
4. Why do you study OOD prinaiple (5marks)

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